



Chapter 4 More on Pandas

# Data Analytics Using Python

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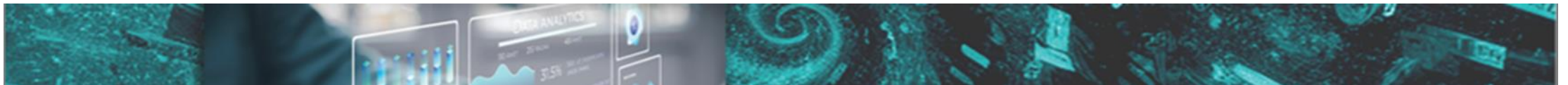


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<https://he.kendallhunt.com/product/data-analytics-using-python>

# Grouping and aggregating data

- Grouping: group values in one or more columns are distributed across different groups
  - `groupby()`: to create subsets of data with one or more columns as group keys
  - need to perform existing or user-defined operations after using `groupby()` such as `mean()`, `max()`, `sum()`, etc
- Aggregation operations, such as `sum()`, `mean()`, and others, can be performed on subsets of data

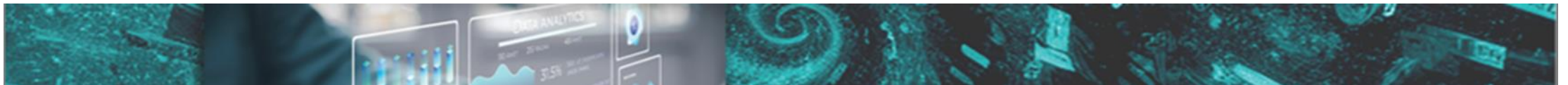




# Merging and Joining Dataframes

- When multiple dataframes have compatible row indexes or columns, they can be combined along either rows (axis=0) or columns (axis=1)

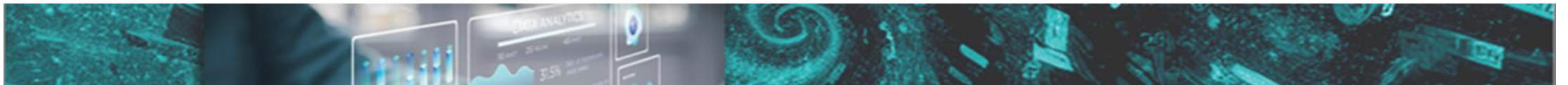
| function        | description                                                                                                                                                  |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>concat()</b> | stack multiple DataFrames vertically or horizontally<br><br>pd.concat(sequence of DataFrames or Series,<br>axis=0,join='outer',ignore_index=False,keys=None) |
| <b>merge()</b>  | combine multiple DataFrames based on matching columns<br><br>pd.merge(left, right, how='inner', on=None, left_index=False, right_index=False)                |
| <b>join()</b>   | combine multiple DataFrames based on matching columns or indexes<br><br>df.join(other_df, on=None, how='left', lsuffix='', rsuffix='')                       |



# Reshaping DataFrames

- reshape a wide DataFrame to a long DataFrame or a long DataFrame to a wide DataFrame

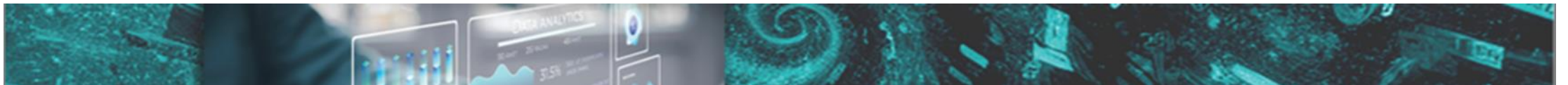
| function                         | description                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>pivot_table()</b>             | Summarize and reshape data<br><code>df.pivot_table(values=None,index=None,columns=None,aggfunc=None)</code>                                    |
| <b>melt()</b>                    | Reshape a DataFrame from long format to wide format<br><code>pd.melt(df,id_vars=None,value_vars=None,var_name=None,value_name='value')</code>  |
| <b>Stack</b><br><b>unstack()</b> | allow DataFrames be reshaped with pivoting the levels of their indexes<br><code>df.stack(level=-1)</code><br><code>df.unstack(level=-1)</code> |



# Working with Time Series

- time series data: specialized date-time related objects

| function               | description                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DatetimeIndex()</b> | Create index for time-based data<br><code>pd.DatetimeIndex(data,freq=None,tz=None)</code>                                                    |
| <b>datetime()</b>      | Create datetime object from datetime module<br><code>datetime.datetime(year,month,day,hour=0,minute=0,second=0)</code>                       |
| <b>date_range()</b>    | Create a DatetimeIndex object with the specified date range<br><code>pd.date_range(start=None,end=None,periods=None,freq='D',tz=None)</code> |
| <b>Grouper()</b>       | Create a Grouper object for groupby() method to do grouping<br><code>pd.Grouper(key=None,level=None,freq=None,axis=0,sort=False)</code>      |



# DataFrame with File Input and Output

- creating DataFrames from CSV files: `read_csv()`
- writing data to a CSV file: `to_csv()`
- reading from Excel files: `read_excel()`
- writing to Excel files: `to_excel()`
- read data from database file: `read_sql()`
- write data to database file: `to_sql()`

